

Project Design Phase-II

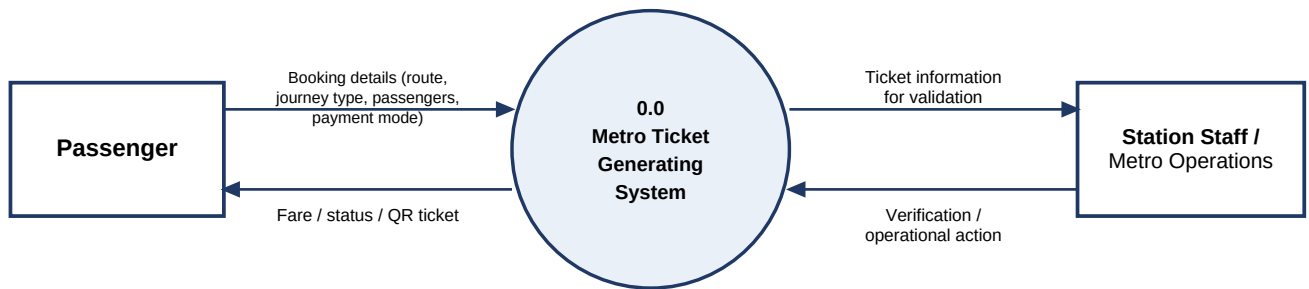
Data Flow Diagram & User Stories

Date	29 September 2026
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	4 Marks

Data Flow Diagrams:

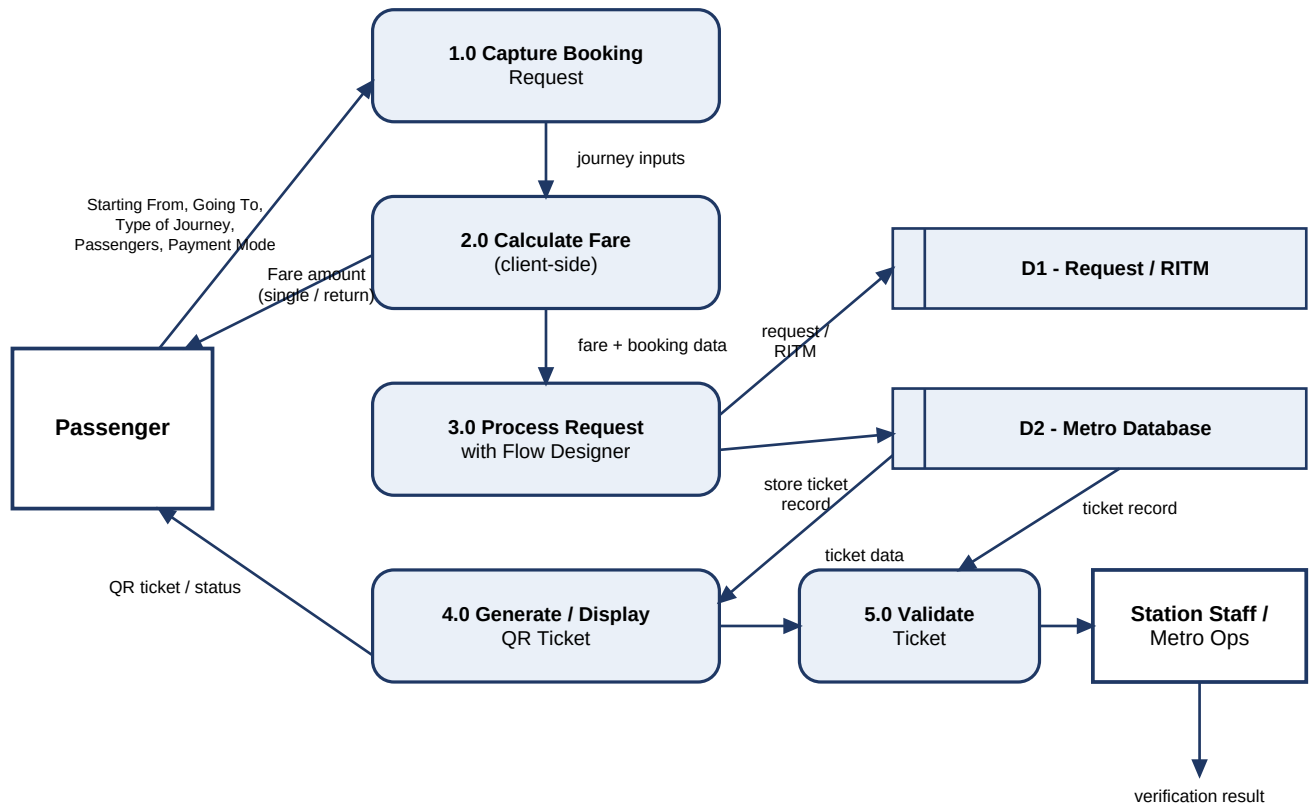
The DFD represents how metro booking information moves through the ServiceNow solution: passenger inputs enter through the Service Catalog, fare information is calculated, Flow Designer processes the request, booking data is stored in the custom Metro Database table (station data is read from the Metro Station's Details table), and QR ticket information is presented for passenger and station-side validation.

DFD Level 0 - Context Diagram



Level 0 treats the Metro Ticket Generating System as one process. The Passenger supplies booking details and receives fare/status/QR output. Station Staff or Metro Operations receive ticket information for validation and may perform operational actions.

DFD Level 1 - Detailed Data Flow



Level 1 Process Description

No.	Process	Data Flow / Function
1	Capture Booking Request	Collect Starting From, Going To, Type of Journey, Number of Passengers and Payment Mode from the Service Catalog. Station options come from the Metro Station's Details table.
2	Calculate Fare	Read the displayed station values on screen, convert them to station positions, derive the number of stops and fare band (Rs. 20 / 30 / 40 / 50), multiply by passenger count, apply 2x for return, and auto-fill the single-journey or including-return amount field. Identical source and destination is blocked.
3	Process Request with Flow Designer	Read catalog variables, execute the automation path, create/update records and trigger required downstream actions.
4	Generate / Display QR Ticket	Create or prepare QR ticket data and render it through a Service Portal-compatible widget / spModal experience.
5	Validate Ticket	Provide ticket information to station staff or operations for verification.

Data Stores

- **D1 - Request / RITM:** Maintains the ServiceNow request and request-item context created when the passenger submits the catalog item.
- **D2 - Metro Database:** Custom structured storage for metro ticket / booking details mapped from the submitted catalog variables.
- **Reference data - Metro Station's Details:** Station master table (station_name) that feeds the Starting From and Going To reference variables.

User Stories

The user stories below translate the implemented and planned functional requirements into user-centered tasks. Priority and release values are aligned with the project's four-phase roadmap rather than claiming features that were not implemented.

User Type	Functional Requirement (Epic)	User Story No.	User Story / Task	Acceptance Criteria	Priority	Release
Passenger	Ticket Booking	USN-1	As a passenger, I can open the metro booking catalog item and enter my journey details.	Starting From, Going To, Type of Journey, Number of Passengers and Payment Mode can be entered successfully.	High	Phase 1
Passenger	Fare Calculation	USN-2	As a passenger, I can view the calculated fare before finalizing the booking.	Correct fare is calculated from the configured station/fare logic, passenger count and journey type, and shown in the relevant amount field.	High	Phase 2
Passenger	Ticket Submission	USN-3	As a passenger, I can submit my metro ticket request through the ServiceNow portal.	A request / RITM is created and the booking enters the configured Flow Designer process.	High	Phase 2
Passenger	Request Tracking	USN-4	As a passenger, I can track the status of my submitted booking request.	The request is visible with an understandable processing status.	High	Phase 3
Passenger	Digital Ticket	USN-5	As a passenger, I can open a QR-based digital ticket after the booking is processed.	The QR/ticket view loads successfully in the portal and displays the booking proof.	High	Phase 3
Station Staff	Ticket Validation	USN-6	As station staff, I can view or verify the passenger's ticket information from the QR ticket.	Ticket details correspond to the stored booking data and can be validated.	High	Phase 3
Metro Operations	Booking Records	USN-7	As an operations user, I can access structured metro booking records for operational use.	Mapped booking fields are stored in the custom Metro Database table.	High	Phase 2
Metro Operations	Workflow Monitoring	USN-8	As an operations user, I can inspect Flow Designer execution details when a booking fails.	Execution details identify the failed step, condition path, input or output involved.	High	Phase 4
Administrator	Access Control	USN-9	As an administrator, I can configure roles and ACLs to control access to ticket records.	Unauthorized users cannot access protected operational data beyond their permitted role.	High	Phase 4
Administrator	Maintainability	USN-10	As an administrator, I can maintain the solution primarily through Flow Designer, UI Policies and limited client-side logic inside the scoped application.	Core workflow can be understood and modified without unnecessary custom server-side code.	Medium	Phase 4
Passenger	Dynamic Form Behaviour	USN-11	As a passenger, I see only the amount field that matches my Type of Journey (single or return).	UI policies show or hide the single-journey and including-return amount fields when the journey type changes.	High	Phase 2

User Type	Functional Requirement (Epic)	User Story No.	User Story / Task	Acceptance Criteria	Priority	Release
Passenger	Route Validation	USN-12	As a passenger, I am prevented from choosing the same station as source and destination.	An error is shown and the invalid selection is blocked until a different station is chosen.	High	Phase 2
Passenger	Email/SMS Ticket	USN-13	As a passenger, I would like to receive my ticket through email or SMS as an additional delivery channel.	Ticket details are delivered through the selected notification channel.	Medium	Future
Passenger	Digital Payment	USN-14	As a passenger, I would like to pay using UPI/cards/wallets during booking.	Successful payment is confirmed and associated with the ticket request.	Medium	Future

Acceptance Focus

The critical acceptance path is: passenger enters valid journey information -> fare is calculated -> request is submitted -> Flow Designer processes and stores the booking -> QR ticket is displayed -> station-side ticket information can be verified. Debugging of fare mismatches, variable mappings and QR rendering is therefore directly linked to these high-priority user stories.